01 plate_recognition

License-plate localization

```
# 1. SoC description (abridged)
"Parking-entrance plate localization:
 one MIPI CSI-2 camera; I2C sensor setup;
 Ethernet masks/regions to parking host."
technology: 28nm; clock target: 75 MHz

# 2. Structured requirements (abridged)
interfaces: CSI-2 pixels / I2C / Ethernet
storage: ping-pong frames / local SRAM
fabric: AXI burst / AXI-Lite control
debug: cycle and transaction counters

# 3. Library lookup (one example)
need: camera pixels + frame markers
category: communication
IP: mipi_csi2
features: pixel stream / frame synchronization

# 4. Exported recommendation (abridged)
selected: [mipi_csi2, i2c, ethernet,
  performance_counter, frame_buffer,
  sram_scratchpad, axi_dma_interconnect,
  axi_lite_ctrl]
unmatched: []
IP area: 1.49 mm2 (limit: 13 mm2)
IP power: 48 mW (limit: 1500 mW)
status: recommendation_only
```

Case 1. An abridged view of the recorded SoC description, structured requirements, one selected library entry, and exported IP recommendations. Resource totals cover reusable IPs only.

Single-column preview: 3.5 in wide; code: 8 pt.

02 recognition

Color-based object recognition

```
# 1. SoC description (abridged)
"Industrial green-marker localization:
 BGR frames arrive over SPI; target
 coordinates are reported over UART."
technology: 28nm; clock target: 75 MHz

# 2. Structured requirements (abridged)
interfaces: SPI slave input / UART output
storage: 320x240 BGR frame / local SRAM
control: AXI-Lite start / done / status
debug: JTAG access

# 3. Library lookup (one example)
need: target coordinates + configuration
category: communication
IP: uart
features: serial / short-distance link

# 4. Exported recommendation (abridged)
selected: [uart, frame_buffer, axi_lite_ctrl]
unmatched: [SPI slave frame ingress, JTAG debug,
  SRAM scratchpad access]
IP area: 0.33 mm2 (limit: 25 mm2)
IP power: 11.5 mW (limit: 1500 mW)
status: recommendation_only
```

Case 2. An abridged view of the recorded SoC description, structured requirements, one selected library entry, and exported IP recommendations. Resource totals cover reusable IPs only.

Single-column preview: 3.5 in wide; code: 8 pt.

03 scan

Document scanning

```
# 1. SoC description (abridged)
"USB document-cleanup device: receive
 RGB pages from a host; return binary
 masks and document bounding boxes."
technology: 28nm; clock target: 75 MHz

# 2. Structured requirements (abridged)
interfaces: USB device / page-transfer DMA
storage: strided page buffers / local SRAM
control: AXI-Lite start / done / status
debug: cycle and transaction counters

# 3. Library lookup (one example)
need: page-processing latency measurement
category: debug
IP: performance_counter
features: cycle / transaction counters

# 4. Exported recommendation (abridged)
selected: [performance_counter, sram_scratchpad,
  axi_lite_ctrl]
unmatched: [USB device controller,
  page-transfer DMA, page/frame buffering]
IP area: 0.55 mm2 (limit: 25 mm2)
IP power: 12.5 mW (limit: 1500 mW)
status: recommendation_only
```

Case 3. An abridged view of the recorded SoC description, structured requirements, one selected library entry, and exported IP recommendations. Resource totals cover reusable IPs only.

Single-column preview: 3.5 in wide; code: 8 pt.

04 wami

Wide-area motion imagery

```
# 1. SoC description (abridged)
"Airborne foreground-detection payload:
 raw CSI-2 camera input, I2C sensor setup,
 Ethernet masks/status to mission host."
technology: 28nm; clock target: 75 MHz

# 2. Structured requirements (abridged)
I/O: raw CSI-2 / I2C / Ethernet
storage: current/reference frames / DDR
fabric: frame-ordered DMA / AXI / SRAM
control/debug: AXI-Lite / event tracing

# 3. Library lookup (one example)
need: camera sensor-register control
category: communication
IP: camera_i2c_ctrl
features: I2C/SCCB / exposure control

# 4. Exported recommendation (abridged)
selected: [camera_i2c_ctrl, ethernet,
  trace_monitor, frame_buffer,
  ddr_memory_controller, axi_dma_interconnect,
  sram_scratchpad, axi_lite_ctrl]
unmatched: [raw MIPI CSI-2 receiver,
  frame-ordered streaming DMA]
IP area: 1.88 mm2 (limit: 25 mm2)
IP power: 59.5 mW (limit: 1500 mW)
status: recommendation_only
```

Case 4. An abridged view of the recorded SoC description, structured requirements, one selected library entry, and exported IP recommendations. Resource totals cover reusable IPs only.

Single-column preview: 3.5 in wide; code: 8 pt.

05 nightvision

Night-vision enhancement

```
# 1. SoC description (abridged)
"Dual-sensor night-viewing terminal:
 visible + IR frames over two CSI-2 inputs;
 SPI panel commands; local fused display."
technology: 28nm; clock target: 75 MHz

# 2. Structured requirements (abridged)
interfaces: 2x CSI-2 / I2C / SPI panel
video: paired frames / local display
fabric: AXI burst / SRAM / AXI-Lite
debug: fusion/display latency counters

# 3. Library lookup (one example)
need: sensor exposure/gain configuration
category: communication
IP: i2c
features: two-wire / multi-slave control

# 4. Exported recommendation (abridged)
selected: [i2c, performance_counter,
  video_output, sram_scratchpad,
  axi_dma_interconnect, axi_lite_ctrl]
unmatched: [dual MIPI CSI-2 reception,
  SPI panel control, paired-frame buffering]
IP area: 1.07 mm2 (limit: 13 mm2)
IP power: 29 mW (limit: 1500 mW)
status: recommendation_only
```

Case 5. An abridged view of the recorded SoC description, structured requirements, one selected library entry, and exported IP recommendations. Resource totals cover reusable IPs only.

Single-column preview: 3.5 in wide; code: 8 pt.